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EX NO: 3a	GENERATION OF PHASE MODULATED (PM) SIGNALS AND CALCULATION OF PHASE DEVIATION USING MATLAB SIMULATION
DATE:	

Objectives:

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To observe and analyze the time-domain characteristics of the message signal, carrier signal, and phase modulated signal through MATLAB simulation.
- iii) To calculate the phase deviation of the PM signal using the phase sensitivity constant and message signal amplitude.
- iv) To determine the modulation index of the PM signal and examine its relationship with phase deviation.
- v) To investigate the effect of varying message signal amplitude on the phase deviation and modulation characteristics of the PM signal.
- vi) To compare PM waveforms obtained for different phase deviation values and analyze their modulation behavior using MATLAB.

PROGRAM (MATLAB CODE)

```
clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
```

```

fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
phase_dev = kp*Am;
% PM Modulation Index
mp = phase_dev;
% Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Carrier Amplitude = %.2f V\n',Ac);
fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('Phase Deviation = %.2f
radians\n',phase_dev); fprintf('PM Modulation Index
= %.2f\n',mp); % Plotting
figure('Name','Phase Modulation and Phase
Deviation',... 'NumberTitle','off');
subplot(2,2,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,c,'LineWidth',1.5);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(t,pm,'LineWidth',1.5);

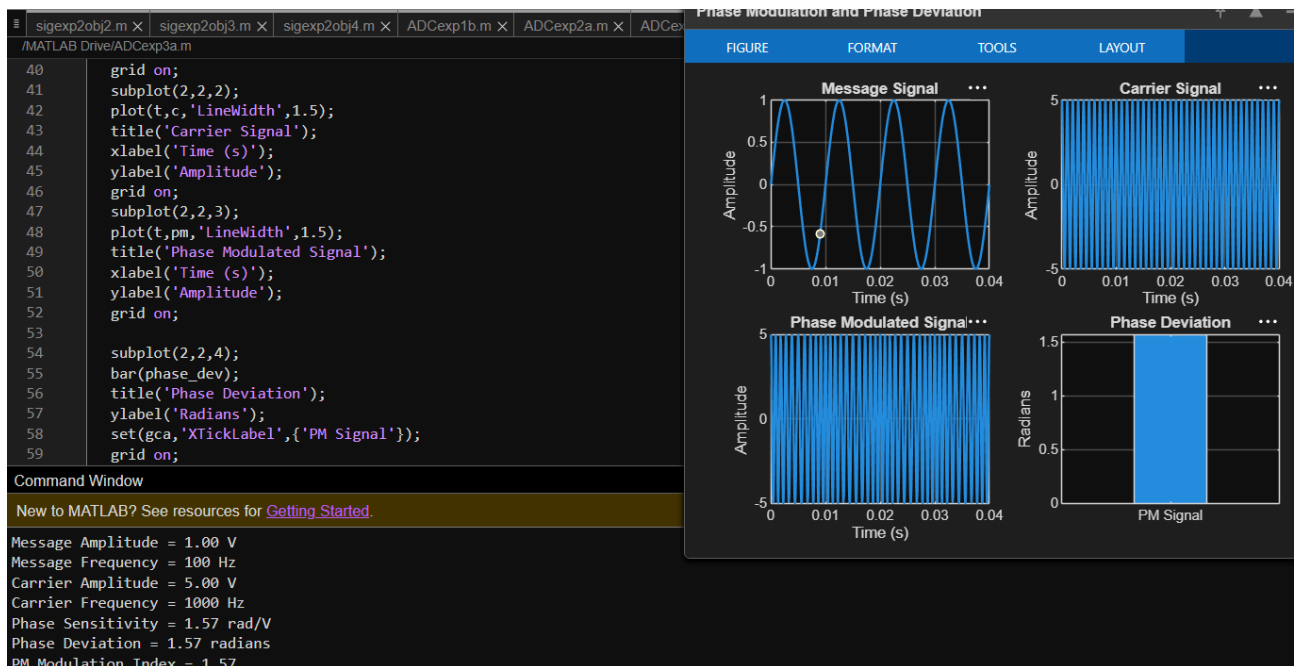
```

```

title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,4);
bar(phase_dev);
title('Phase Deviation');
ylabel('Radians');
set(gca,'XTickLabel',{'PM Signal'});
grid on;

```

OUTPUT (GRAPH)



RESULT

The Phase Modulated (PM) signal was successfully generated and analyzed using MATLAB. The phase deviation was calculated from the phase sensitivity constant and message amplitude.

- i) Message Frequency = **100 Hz**
- ii) Carrier Frequency = **1000 Hz**
- iii) Phase Sensitivity = **1.57 rad/V**
- iv) Phase Deviation = **1.57 radians**
- v) PM Modulation Index = **1.57**
- vi) The PM waveform exhibited phase variations proportional to the message signal amplitude, verifying the characteristics of Phase Modulation.

EX NO: 3b	POWER SPECTRAL DENSITY (PSD) ANALYSIS OF PHASE MODULATED (PM) SIGNALS USING MATLAB
DATE:	

Objectives

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To compute the frequency spectrum of the PM signal using the Fast Fourier Transform (FFT).
- iii) To calculate and plot the Power Spectral Density (PSD) of the PM signal using MATLAB.
- iv) To identify the carrier component and sideband frequencies present in the PM spectrum.
- v) To analyze the effect of phase modulation on the spectral distribution and bandwidth of the transmitted signal.
- vi) To investigate the influence of modulation index on the Power Spectral Density characteristics of the PM signal.

PROGRAM (MATLAB CODE)

```

clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
beta = kp*Am;
% FFT

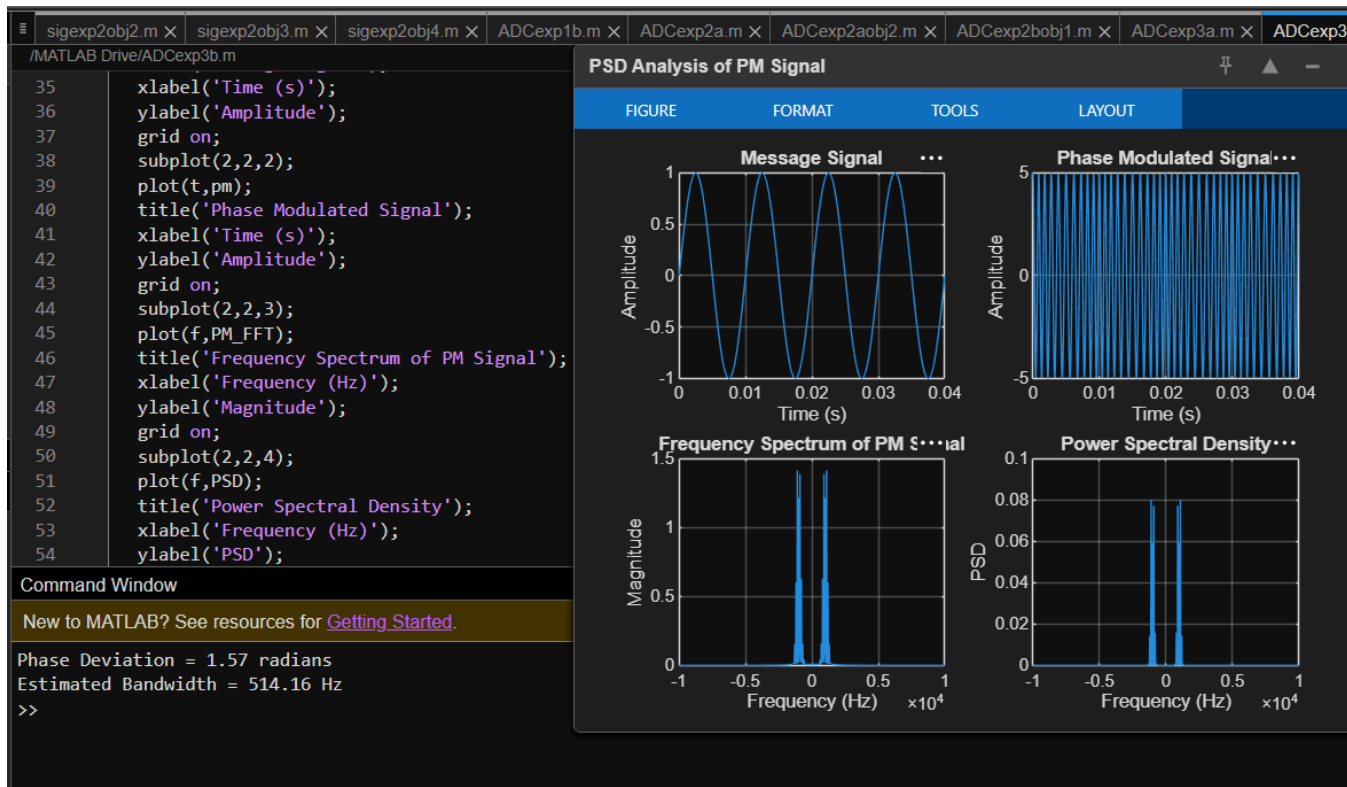
```

```

N = length(pm);
f = linspace(-fs/2,fs/2,N);
PM_FFT = abs(fftshift(fft(pm)))/N;
% Power Spectral Density
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);
% Approximate Bandwidth
BW = 2*(beta+1)*fm;
fprintf('Phase Deviation = %.2f radians\n',beta);
fprintf('Estimated Bandwidth = %.2f Hz\n',BW);
% Plotting
figure('Name','PSD Analysis of PM Signal','NumberTitle','off');
subplot(2,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(f,PM_FFT);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(2,2,4);
plot(f,PSD);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD');
grid on;

```

OUTPUT (GRAPH)



RESULT

The Power Spectral Density (PSD) of the Phase Modulated (PM) signal was successfully obtained and analyzed using MATLAB.

- Message Frequency = **100 Hz**
- Carrier Frequency = **1000 Hz**
- Phase Deviation = **1.57 radians**
- PM Modulation Index = **1.57**
- Estimated Bandwidth = **514 Hz**
- The PSD plot showed the carrier component and multiple sidebands around the carrier frequency, confirming the frequency-domain characteristics of Phase Modulation.

